

Bartlett's Test

Suppose that the data from population i have a variance-covariance matrix Σ_i , for population $i = 1, 2$. What we wish to do is to test the null hypothesis that Σ_1 is equal to Σ_2 against the general alternative that they are not equal as shown below:

$$H_0 : \Sigma_1 = \Sigma_2 \quad \text{against} \quad H_a : \Sigma_1 \neq \Sigma_2 \quad (1)$$

Here, the alternative is that the variance-covariance matrices differ in at least one of their elements.

The test statistic for Bartlett's Test is given by L -prime as shown below:

$$L' = c\{(n_1 + n_2 - 2)\log|S_p| - (n_1 - 1)\log|S_1| - (n_2 - 1)\log|S_2|\} \quad (2)$$

This involves a finite population correction factor c , which is given below. Inside the brackets above, we have the determinants of the sample variance-covariance matrices for the individual populations as well as the pooled sample variance-covariance matrix. So, inside the brackets is the total number of observations minus 2 times the log of the determinant of the pooled variance-covariance matrix, minus $n_1 - 1$ times the log of the determinant of the sample variance-covariance matrix for the first population, minus $n_2 - 1$ times the log of the determinant of the sample variance-covariance matrix for the second population. (Note that in this formula, the logs are all the natural logs)

The finite population correction factor, c , is given below:

$$c = 1 - \frac{2p^2 + 3p - 1}{6(p + 1)} \left\{ \frac{1}{n_1 - 1} + \frac{1}{n_2 - 1} - \frac{1}{n_1 + n_2 - 2} \right\} \quad (3)$$

It is a function of the number of variables p , and the sample sizes n_1 and n_2 .

Under the null hypothesis, $H_0 : \Sigma_1 = \Sigma_2$, Bartlett's test is approximately chi-square distributed with $p(p+1)$ divided by 2 degrees of freedom. That is,

$$L' \sim \chi^2_{\frac{p(p+1)}{2}} \quad (4)$$

This number of degrees of freedom is equal to the number of unique elements in the variance-covariance matrix (taking into account that this matrix is symmetric). We will reject H_0 at level α if the test statistic exceeds the critical value from the chi-square table evaluated at level α .

$$L' > \chi^2_{\frac{p(p+1)}{2}, \alpha} \quad (5)$$